SEABORN ADVANCED GRAPH

In [ ]:
```
#pip install --upgrade seaborn
```

In [1]:
```
import warnings
warnings.filterwarnings("ignore", category=FutureWarning)
```

In [2]:
```
import seaborn as sns
```

In [3]:
```
sns.get_dataset_names()
```

Out[3]:
```
['anagrams',
 'anscombe',
 'attention',
 'brain_networks',
 'car_crashes',
 'diamonds',
 'dots',
 'dowjones',
 'exercise',
 'flights',
 'fmri',
 'geyser',
 'glue',
 'healthexp',
 'iris',
 'mpg',
 'penguins',
 'planets',
 'seaice',
 'taxis',
 'tips',
 'titanic']
```

In [4]:
```
tips = sns.load_dataset("tips")
tips.head()
```

Out[4]:

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

In [5]:
```
titanic = sns.load_dataset("titanic")
titanic.head()
```

Out[5]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adul
0	0	3	male	22.0	1	0	7.2500	S	Third	man	
1	1	1	female	38.0	1	0	71.2833	C	First	woman	
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	
3	1	1	female	35.0	1	0	53.1000	S	First	woman	
4	0	3	male	35.0	0	0	8.0500	S	Third	man	

In [6]:
```
sns.set_theme(style="darkgrid")
```

In [7]:
```
tips.to_csv("tips_dataset.csv", index=False)
import pandas as pd
```

In [8]:
```
import os
os.getcwd()
```

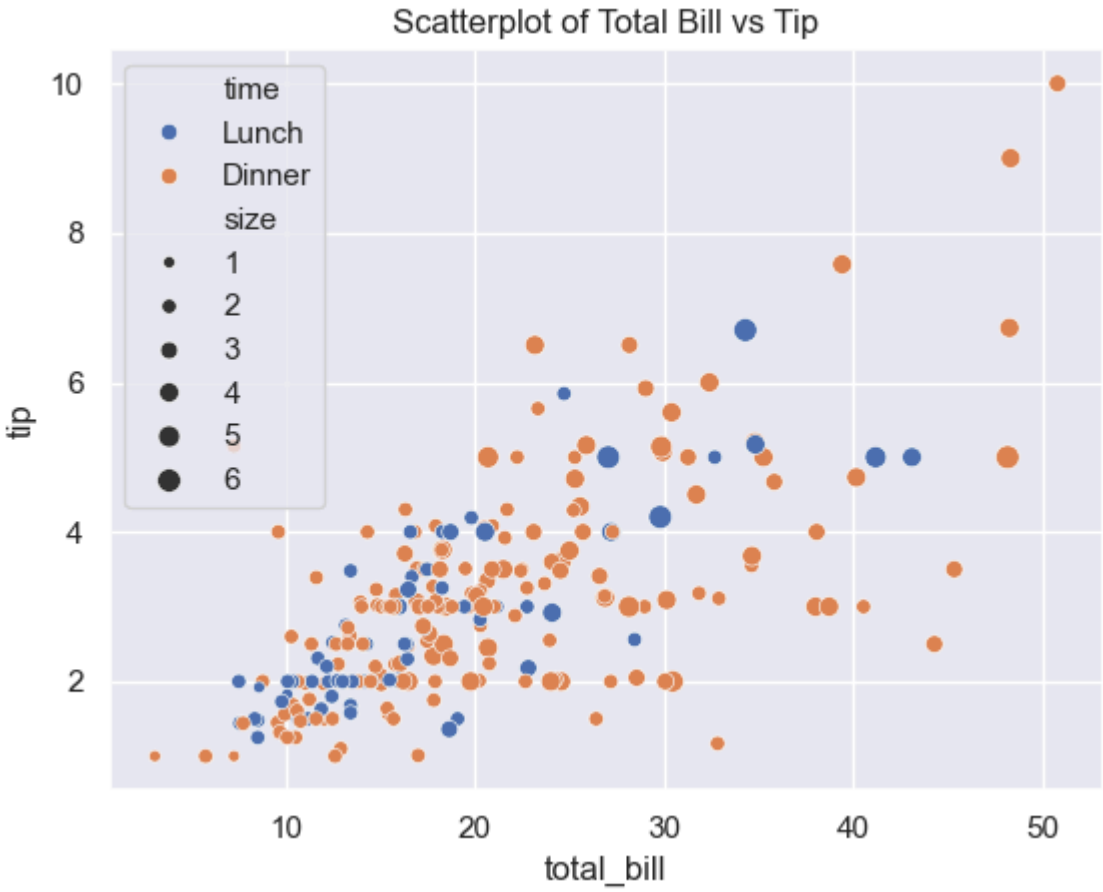
Out[8]:
```
'C:\\Users\\lenovo\\VS_CODE_PROJECT'
```

In [9]:
```
import matplotlib.pyplot as plt
```

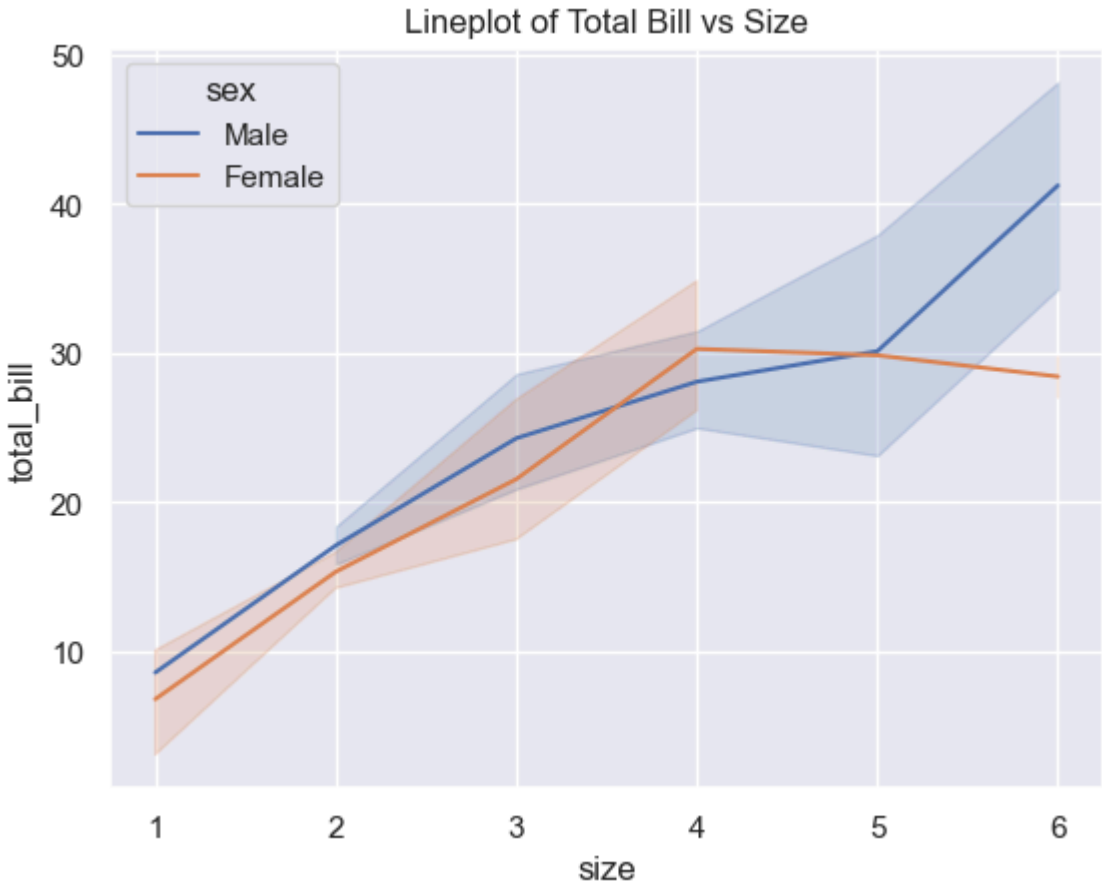
In [11]:
```
plt.figure(figsize=(8, 6))
```

Out[11]:
```
<Figure size 800x600 with 0 Axes>
<Figure size 800x600 with 0 Axes>
```

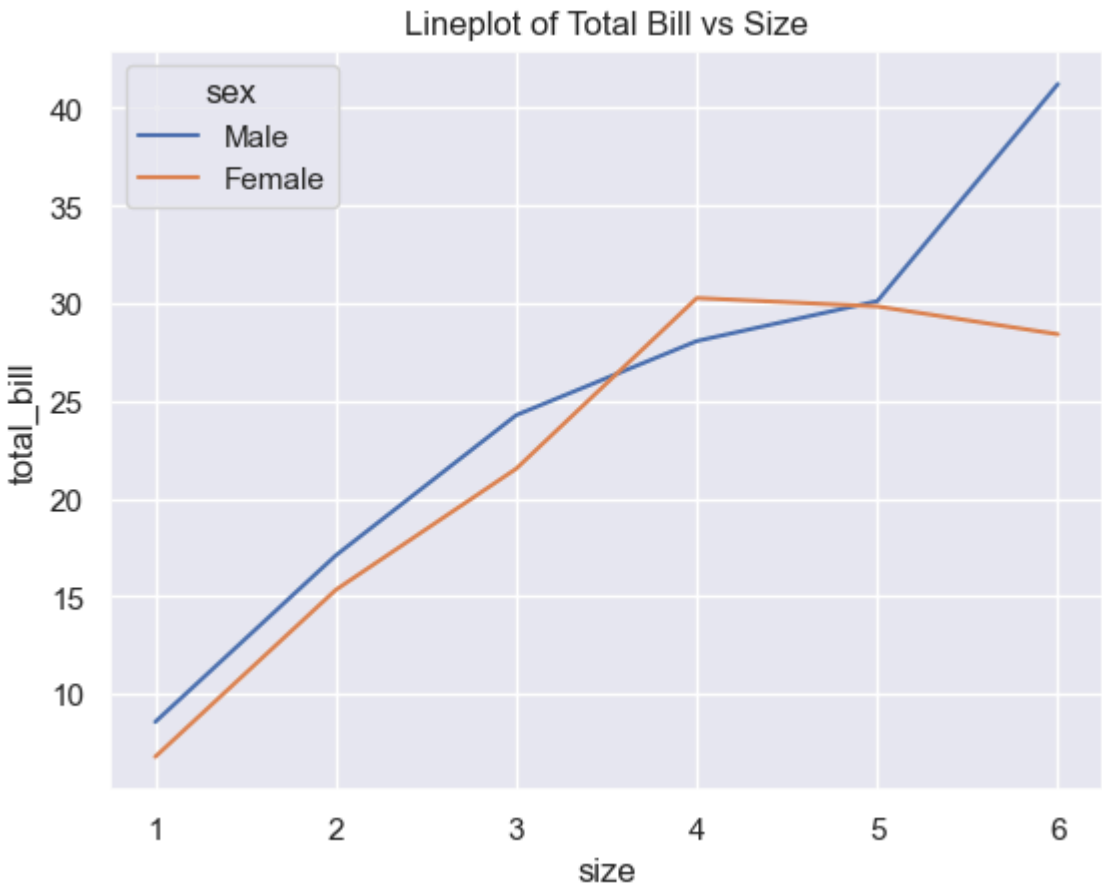
In [12]:
```
# 1. scatter plot

sns.scatterplot(data=tips, x="total_bill", y="tip",hue="time", size="size", pale
plt.title("Scatterplot of Total Bill vs Tip")
plt.show()
```



In [13]:
```
# line plot

sns.lineplot(data=tips, x= 'size', y='total_bill', hue='sex',markers='o')
plt.title("Lineplot of Total Bill vs Size")
plt.show()
```



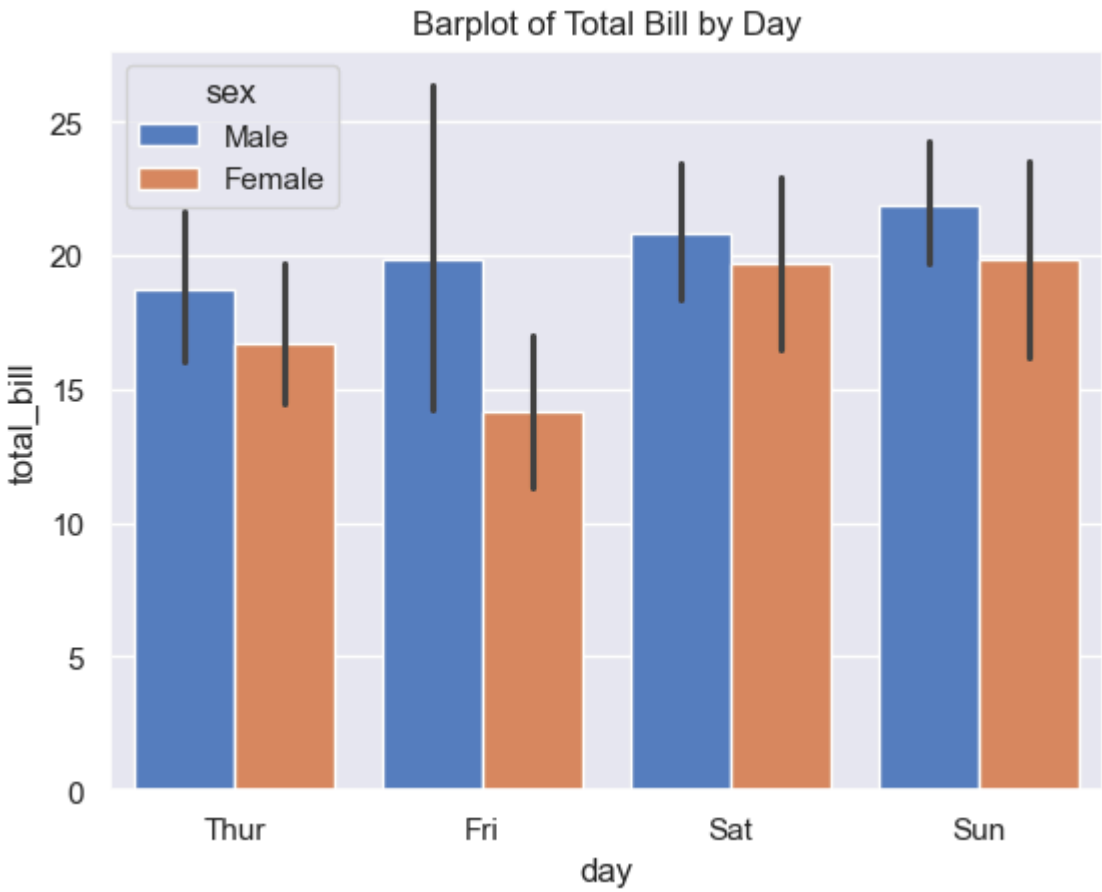
In [17]:
```
sns.lineplot(data=tips, x= 'size', y='total_bill', hue='sex',ci=None, markers='0
plt.title("Lineplot of Total Bill vs Size")
plt.show()
```



In [15]:
```
tips.columns
```

Out[15]:
```
Index(['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size'], dtype='obj
ect')
```

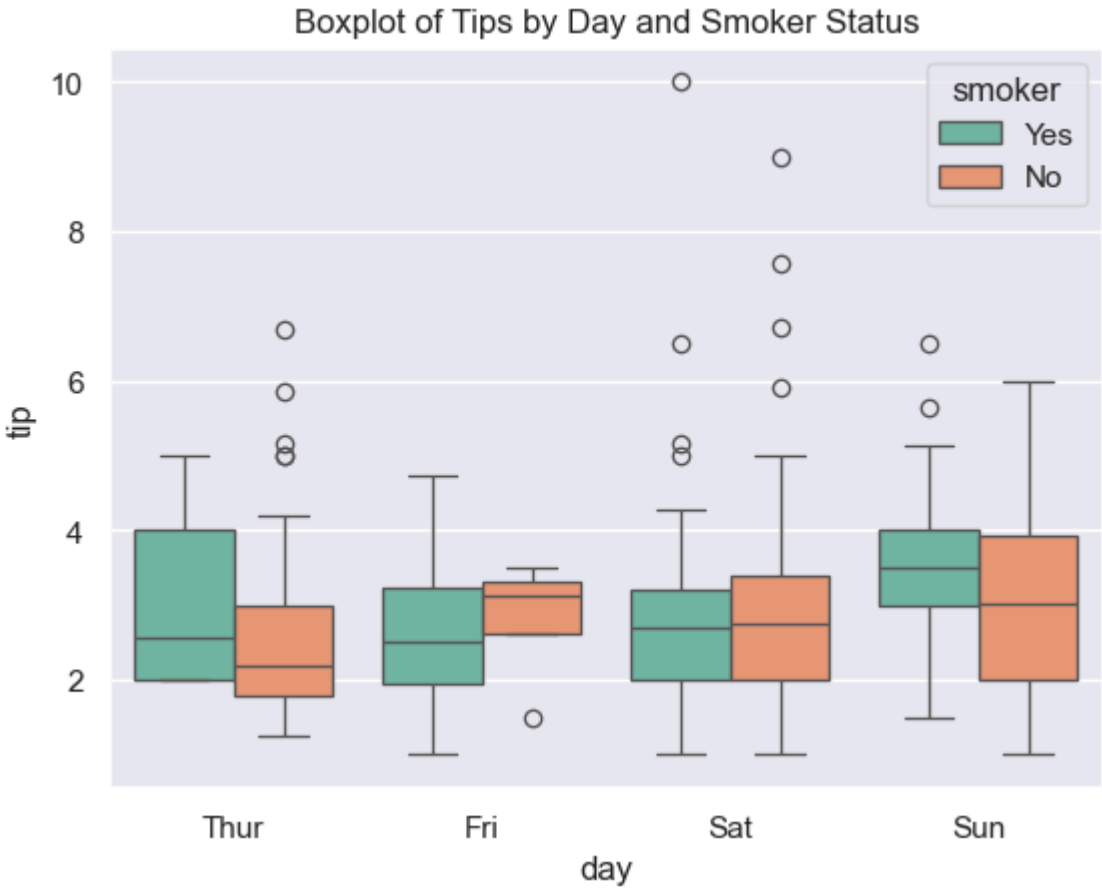
In [18]:
```
#3. bar plot

sns.barplot(data=tips, x='day', y='total_bill', hue = 'sex',palette='muted')
plt.title("Barplot of Total Bill by Day")
plt.show()
```



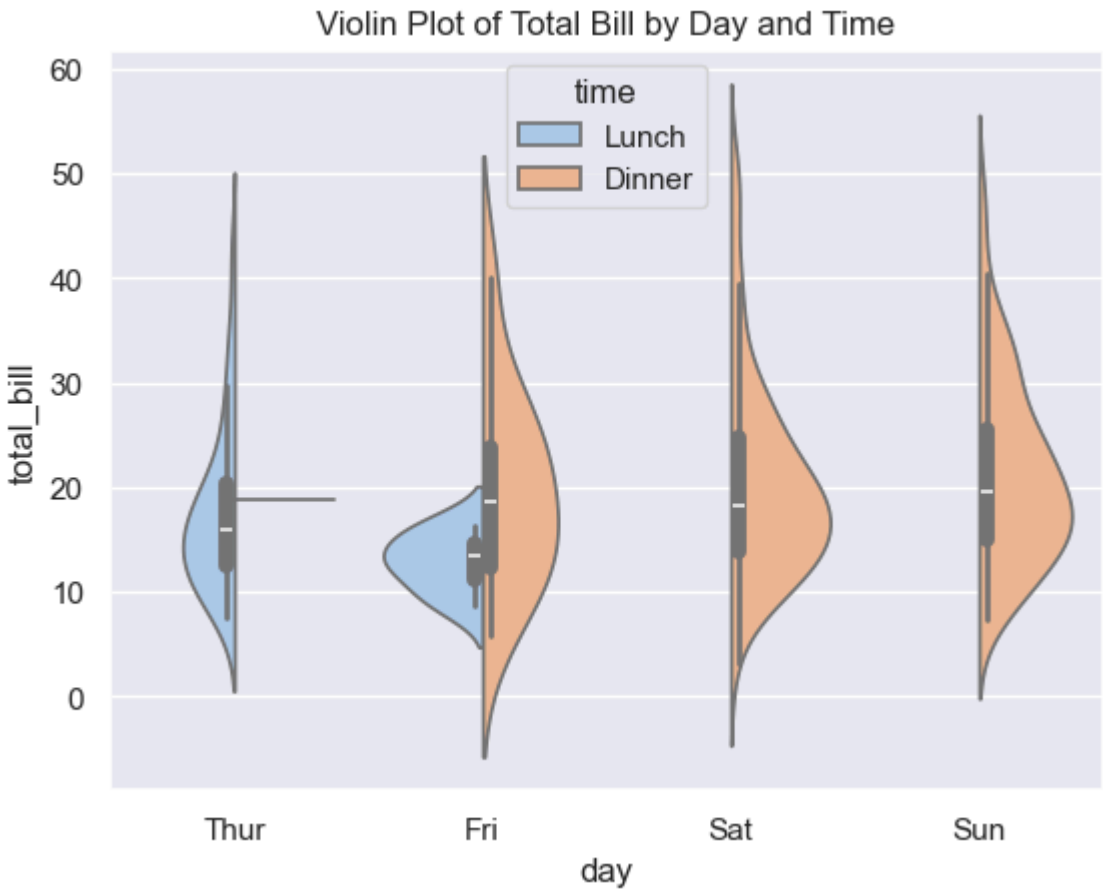
In [19]:
```
tips.columns
```

Out[19]:
```
Index(['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size'], dtype='obj
ect')
```

In [20]:
```
# 4. Boxplot

sns.boxplot(data=tips, x='day', y='tip', hue='smoker', palette='Set2')
plt.title("Boxplot of Tips by Day and Smoker Status")
plt.show()
```



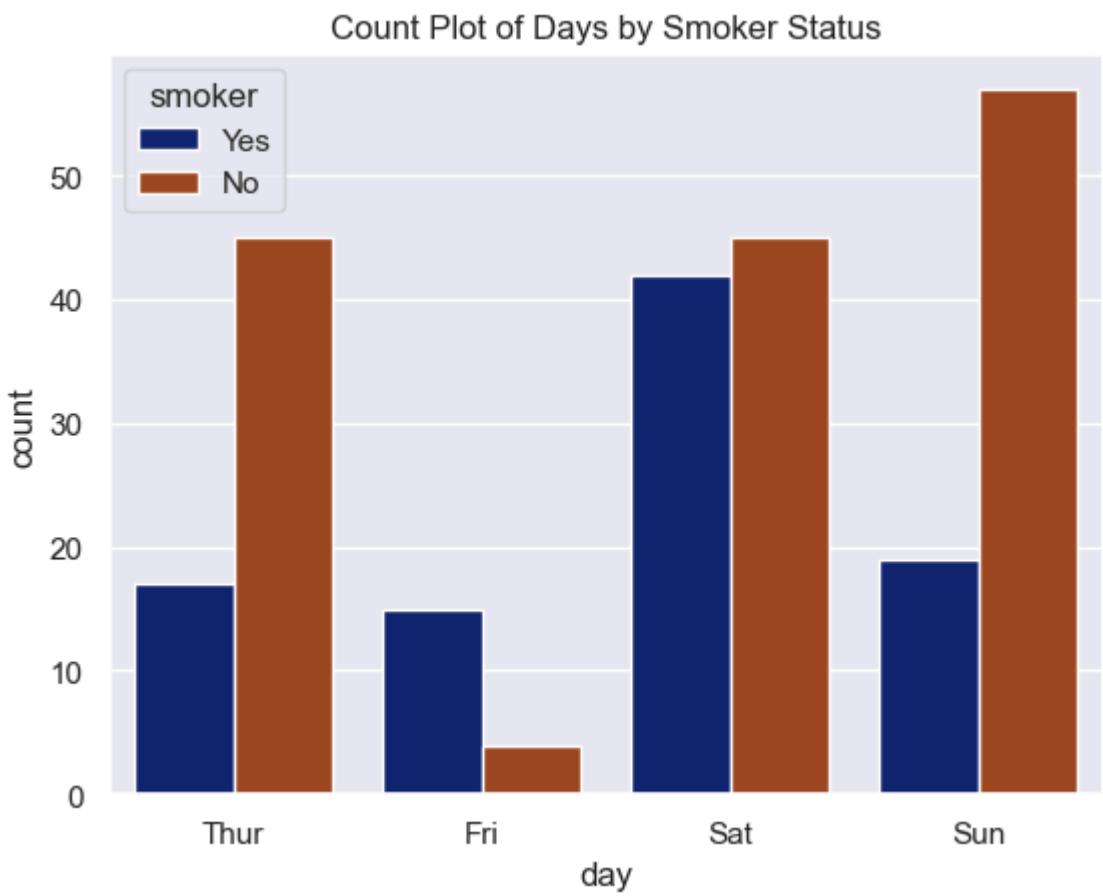
In [21]:
```
# 5. violin plot

sns.violinplot(data=tips, x='day', y='total_bill', hue='time', split=True, palet
plt.title("Violin Plot of Total Bill by Day and Time")
plt.show()
```



In [22]:
```
tips.columns
```

Out[22]:
```
Index(['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size'], dtype='obj
ect')
```

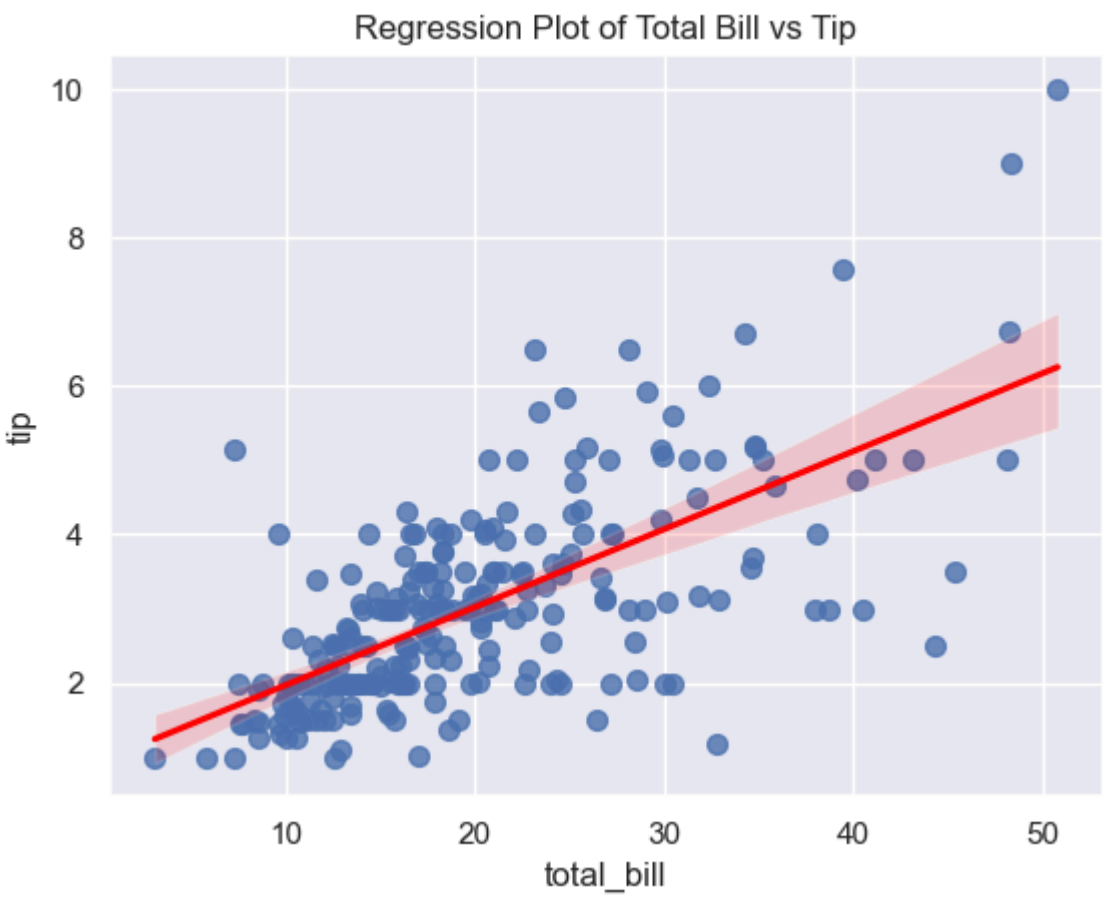
In [23]:
```
#6. count plot

sns.countplot(data=tips, x='day', hue='smoker', palette='dark')
plt.title("Count Plot of Days by Smoker Status")
plt.show()
```



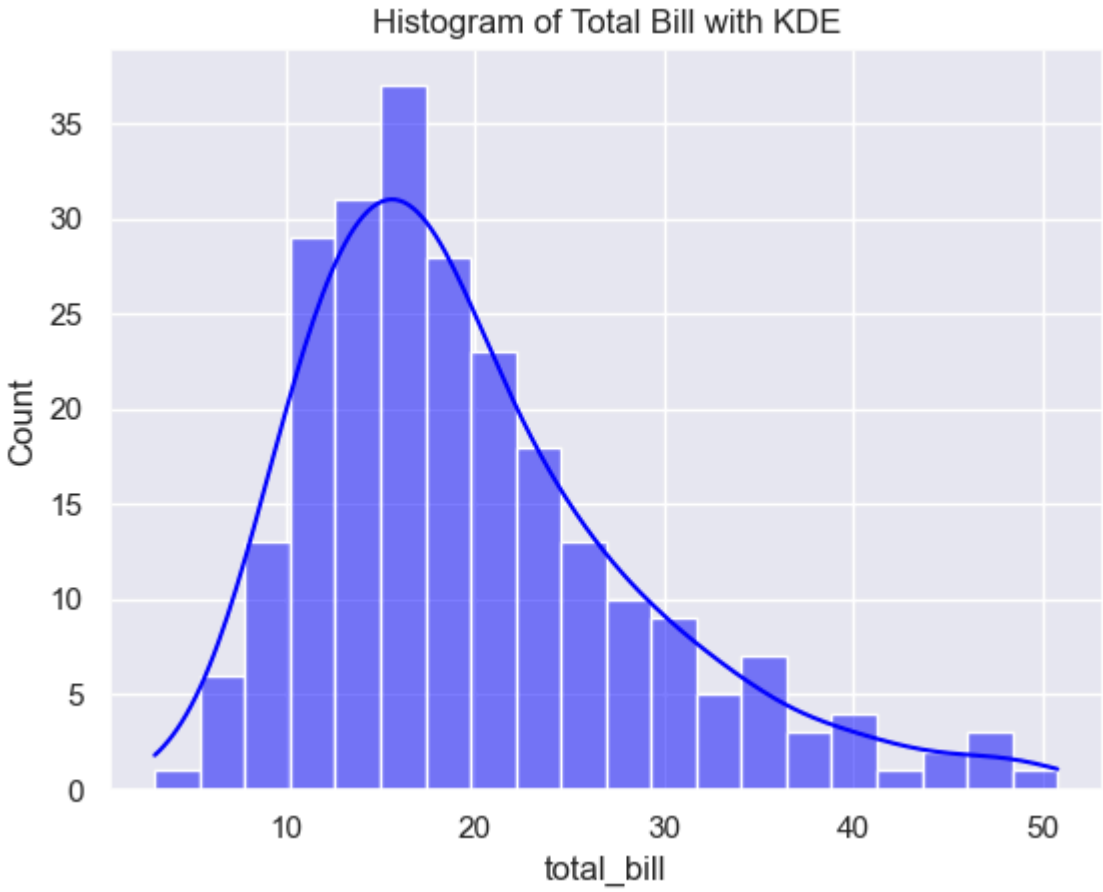
In [24]:
```
tips.columns
```

Out[24]:
```
Index(['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size'], dtype='obj
ect')
```

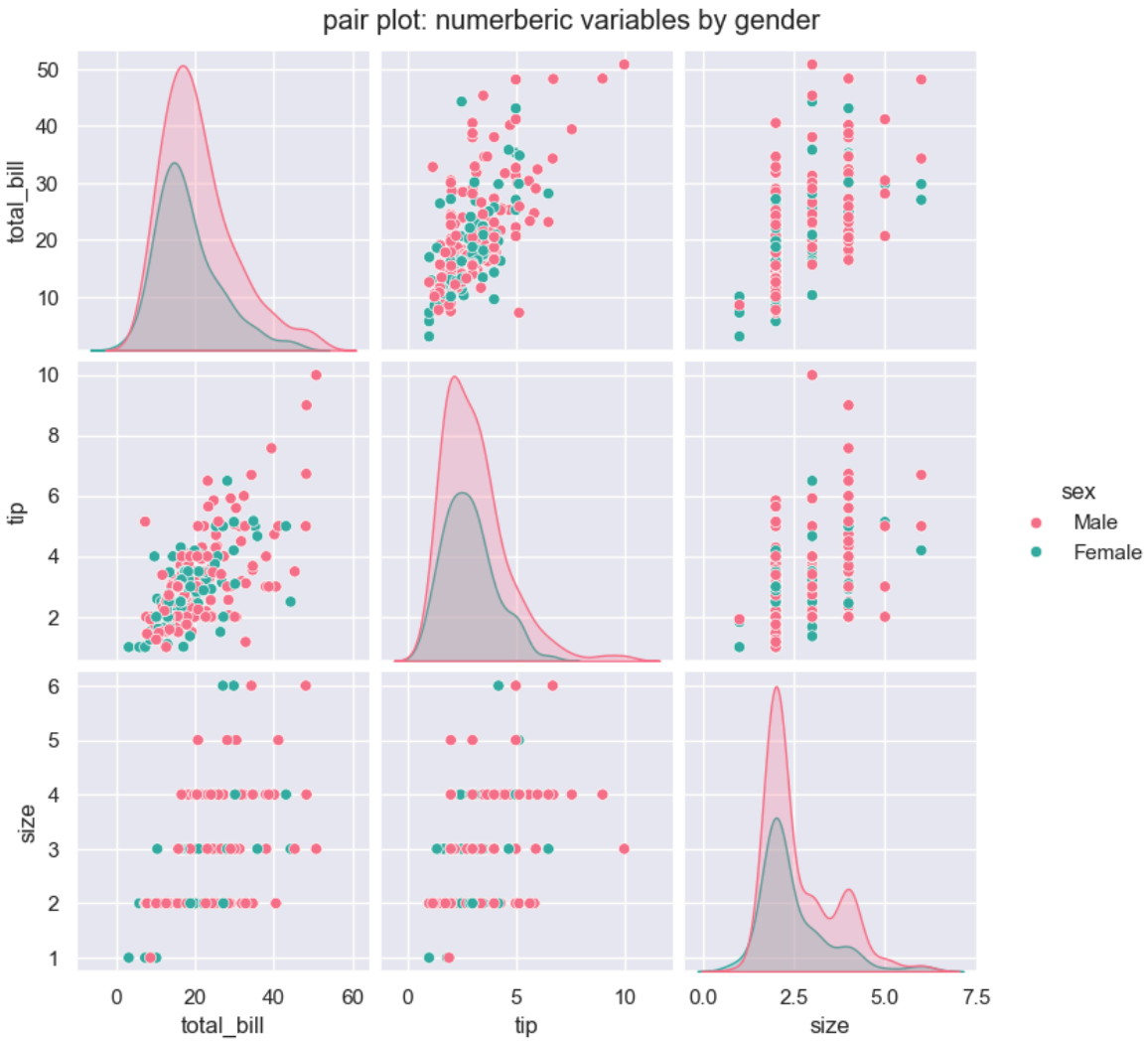
In [25]:
```
#7. regression plot

sns.regplot(data=tips, x='total_bill', y='tip', scatter_kws={'s':50}, line_kws={
plt.title("Regression Plot of Total Bill vs Tip")
plt.show()
```



In [26]:
```
# 8. Histogram of total bill with KDE

sns.histplot(data=tips, x='total_bill', bins=20, kde=True, color='blue')
plt.title("Histogram of Total Bill with KDE")
plt.show()
```

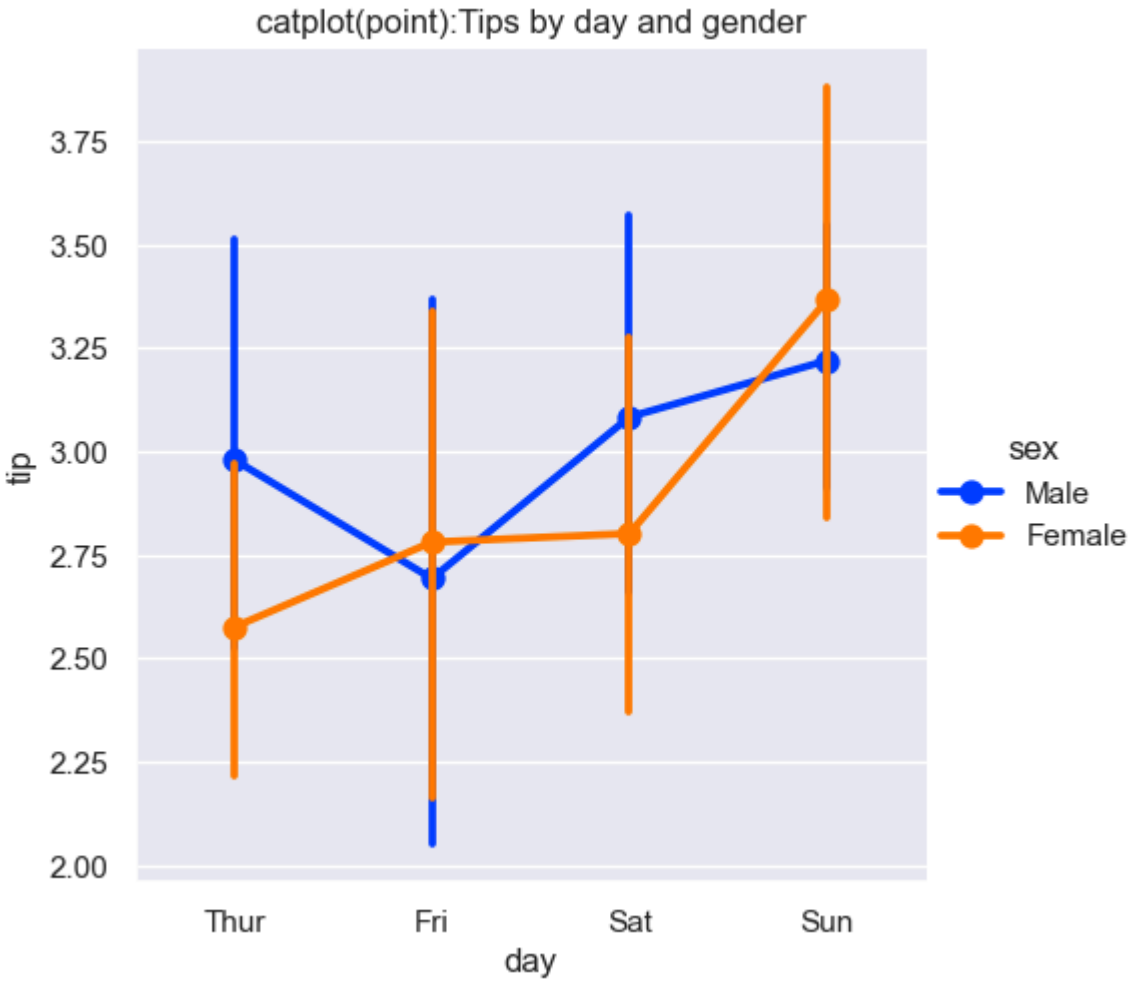


In [27]:
```
#9. pairplot

sns.pairplot(tips, hue='sex', vars=["total_bill", "tip", "size"], palette='husl'
plt.suptitle("pair plot: numerberic variables by gender", y=1.02)
plt.show()
```

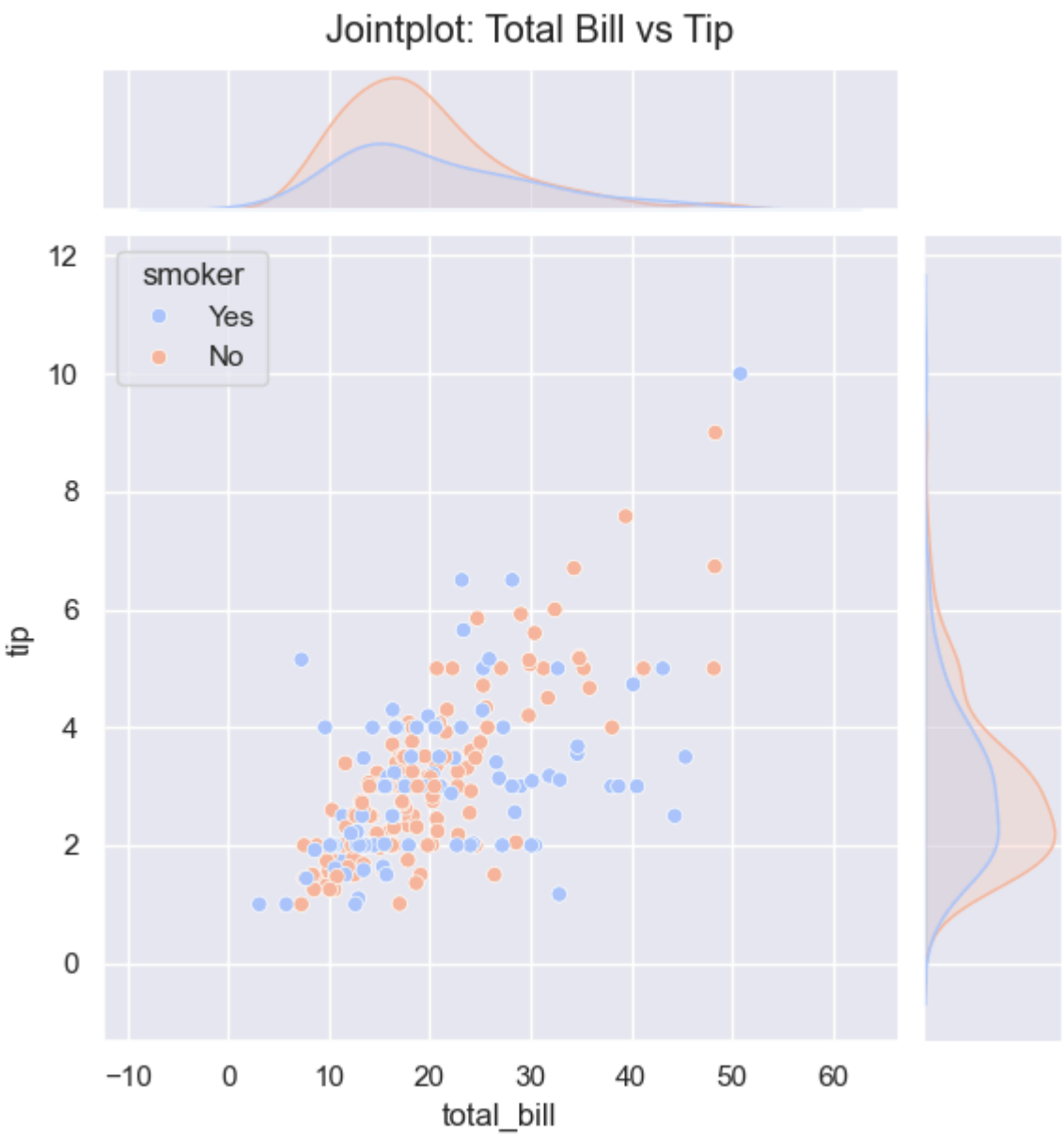


In [ ]:
```
# For 2 attribute we use x-axis and y-axis . what if we have 10 attribute?
# Then comes Heat Map
```

In [28]:
```
# 10 catplot

sns.catplot(data=tips, x='day', y='tip', hue='sex', kind='point', palette='brigh
plt.title("catplot(point):Tips by day and gender")
plt.show()
```

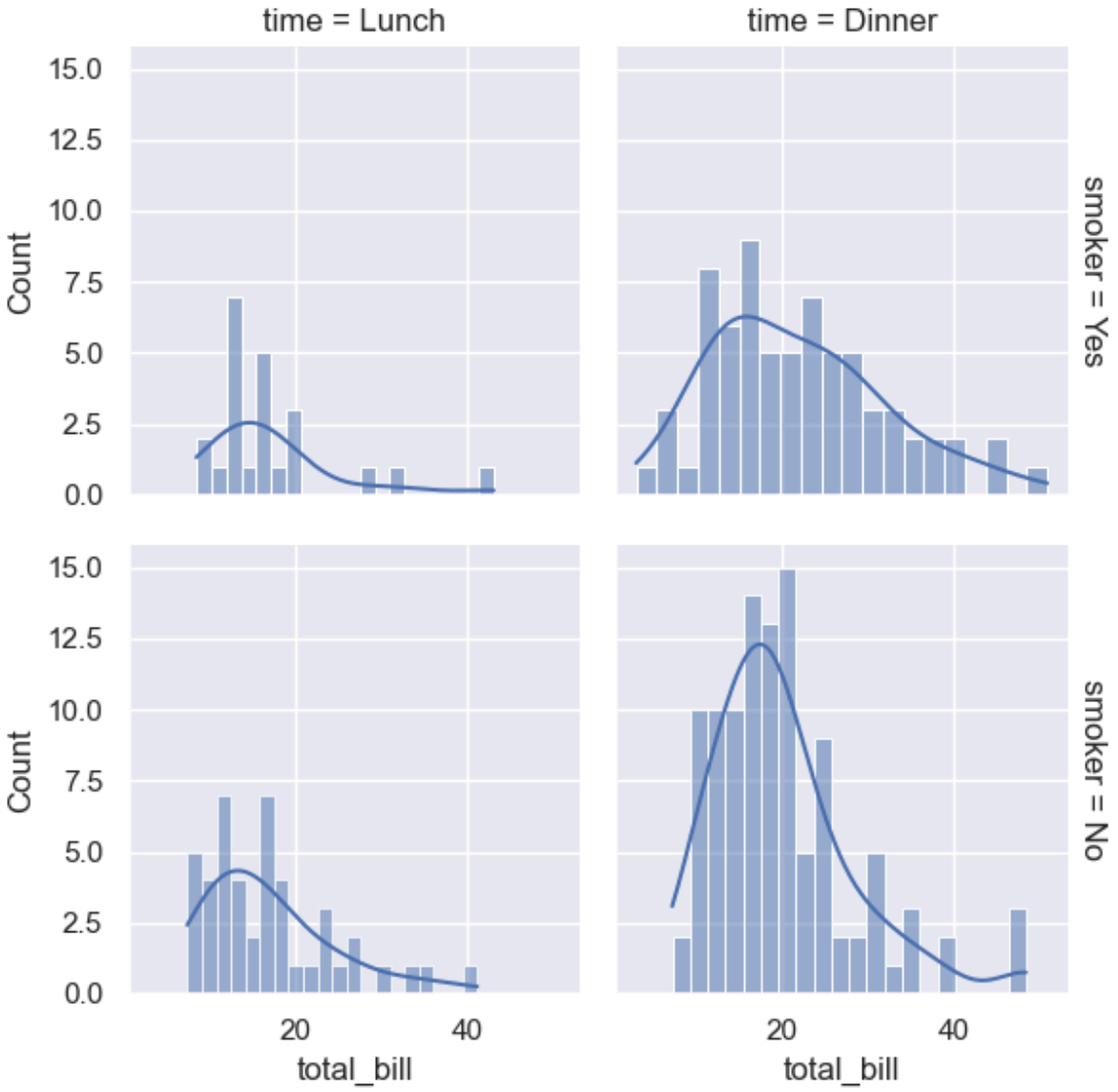


In [29]:
```
# 11. jointplot

sns.jointplot(data=tips, x='total_bill', y='tip', kind='scatter', hue='smoker',
plt.suptitle("Jointplot: Total Bill vs Tip", y=1.02)
plt.show()
```

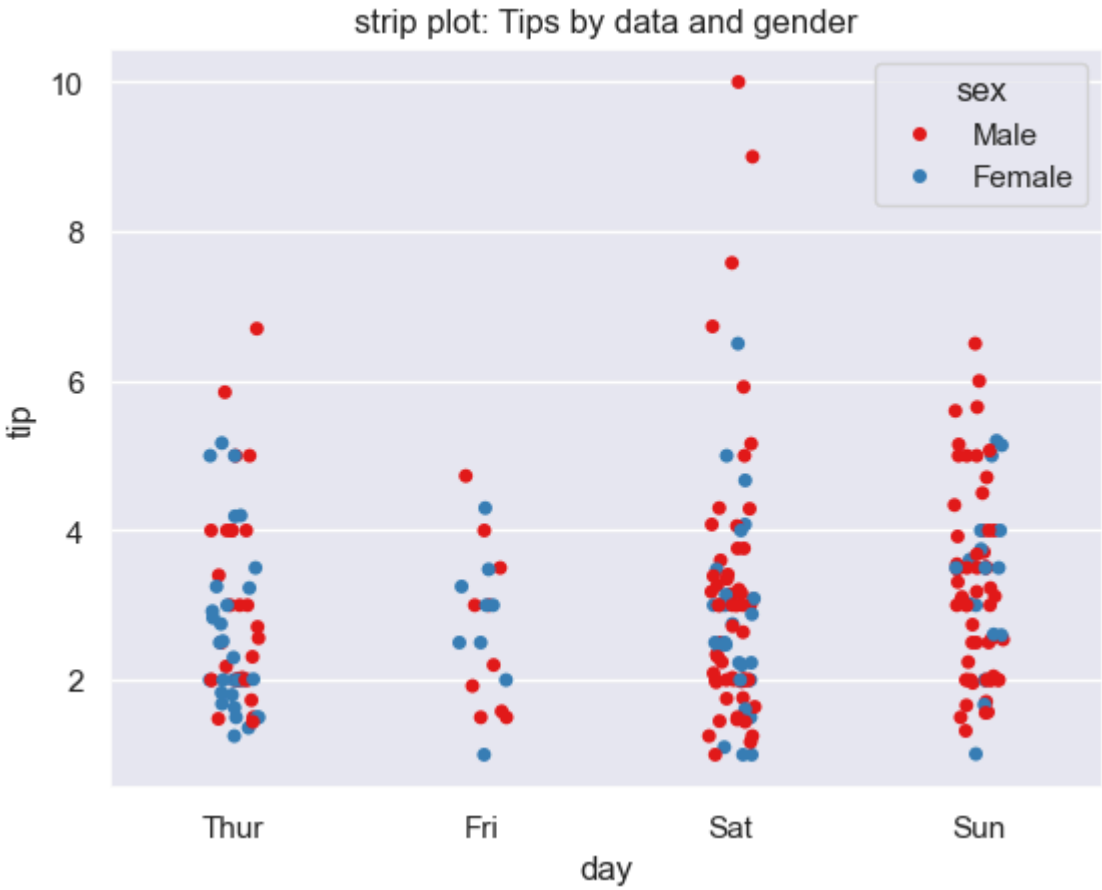


In [30]:
```
# Facetgrid

g = sns.FacetGrid(tips, col='time', row='smoker', margin_titles=True).map(sns.hi
g
```

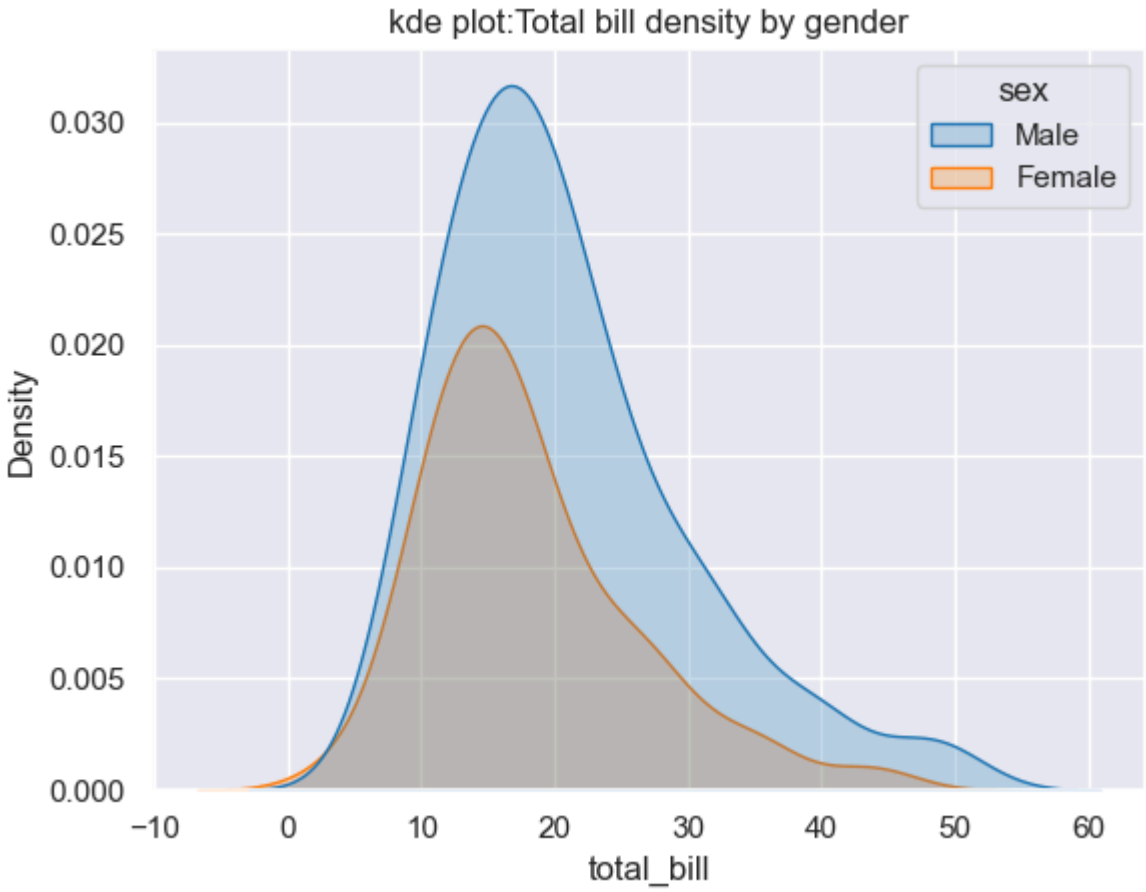
Out[30]:
```
<seaborn.axisgrid.FacetGrid at 0x186c50c39d0>
```



In [31]:
```
# 13. strip plot

sns.stripplot(data=tips, x='day', y='tip', hue='sex', jitter=True, palette='Set1
plt.title("strip plot: Tips by data and gender")
plt.show()
```



In [32]:
```
# 14. KDE PLOT- kernel density plot

sns.kdeplot(data=tips, x='total_bill',hue='sex', fill=True, palette='tab10')
plt.title("kde plot:Total bill density by gender")
plt.show()
```



In [ ]:
```
# Data anlayst we developed graphs
# i have represetn these graph to my mangaer, leader, client
# also i should deliver and i should explain code should not rquired
```